

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

CLASS

INDEX
NUMBER

PRELIMINARY EXAMINATION 2023 SECONDARY 4 EXPRESS / 5 NORMAL ACADEMIC

ADDITIONAL MATHEMATICS

Paper 2

4049/02

28 August 2023

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Calculator Model

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 90.

Total Marks

Mathematical Formulae**1. ALGEBRA***Quadratic Equation*

For the equation $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$.

2. TRIGONOMETRY*Identities*

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area of } \Delta = \frac{1}{2} ab \sin C$$

Answer **all** the questions.

- 1 (a) Differentiate $2e^{2x+1}(\sin 4x)$ with respect to x . [3]

- (b) A curve has equation $y = \frac{x-2}{3x+2}$, where $x \neq -\frac{2}{3}$.
Explain why the curve has no turning points. [3]

- 2 The equation of a circle C_1 , with centre M , is $x^2 + y^2 + px + \left(\frac{p}{2} + 4\right)y + k = 0$, where p and k are constants. It is given that M lies on the line $y = \frac{3}{2}x - 4$.

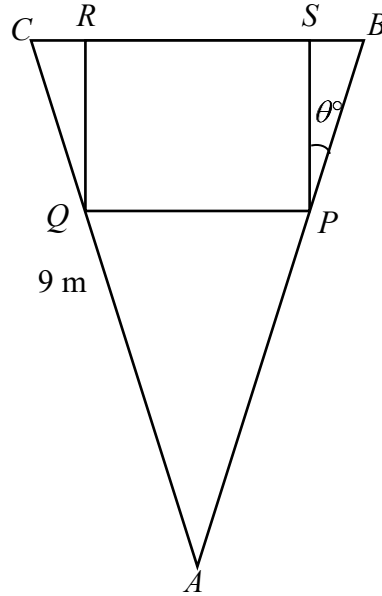
(a) Show that $p = -4$ and find the coordinates of M .

[5]

- (b) Find the value of k , given that $x = -8$ is a tangent to the circle. [2]

- (c) The coordinates of point A is $(14, 8)$ and XY is the diameter of C_1 .
Prove that angle XAY is less than 90° . [2]

- 3 A gardener has a triangular plot of land ABC with a rectangular flower bed $PQRS$ inside. He wants to put up fencing around the perimeter of his triangular land ABC and also along PS , QP and QR . It is given that $AB = AC = 9 \text{ m}$, $AP = 2PB$ and angle $BPS = \theta^\circ$.



- (a) Show that $L \text{ m}$, the length of fencing needed can be expressed in the form $k + 30 \sin \theta + 6 \cos \theta$, where k is a constant.

[3]

- (b) Hence show that L can be expressed in the form of $k + R \sin(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. [3]

- (c) Find the value of θ for which $L = 35$. [3]

- 4 (a) The function $y = \ln\left(\frac{\sqrt{x^2+9}}{2x-4}\right)$ is defined for $x > k$, where k is a constant.
Find the value of k . [2]

- (b) By considering $\frac{dy}{dx}$, explain whether y is an increasing or decreasing function for $x > k$. [5]

- 5 The curve $y = c - a \cos bx$, where a , b and c are positive constants, is defined for $0^\circ \leq x \leq 360^\circ$. The maximum and minimum values of the curve are 9 and -3 respectively. The curve has a period of 180° .

(a) State the values of a , of b and of c .

[3]

(b) Sketch the graph of $y = c - a \cos bx$ for $0^\circ \leq x \leq 360^\circ$.

[3]



- 6 The volume of a prism, $V \text{ cm}^3$, is given by the formula, $V = x(a - bx)$, where $x \text{ cm}$ is the height of the prism and $(a - bx) \text{ cm}^2$ is its base area.

The table below shows corresponding values of x and V .

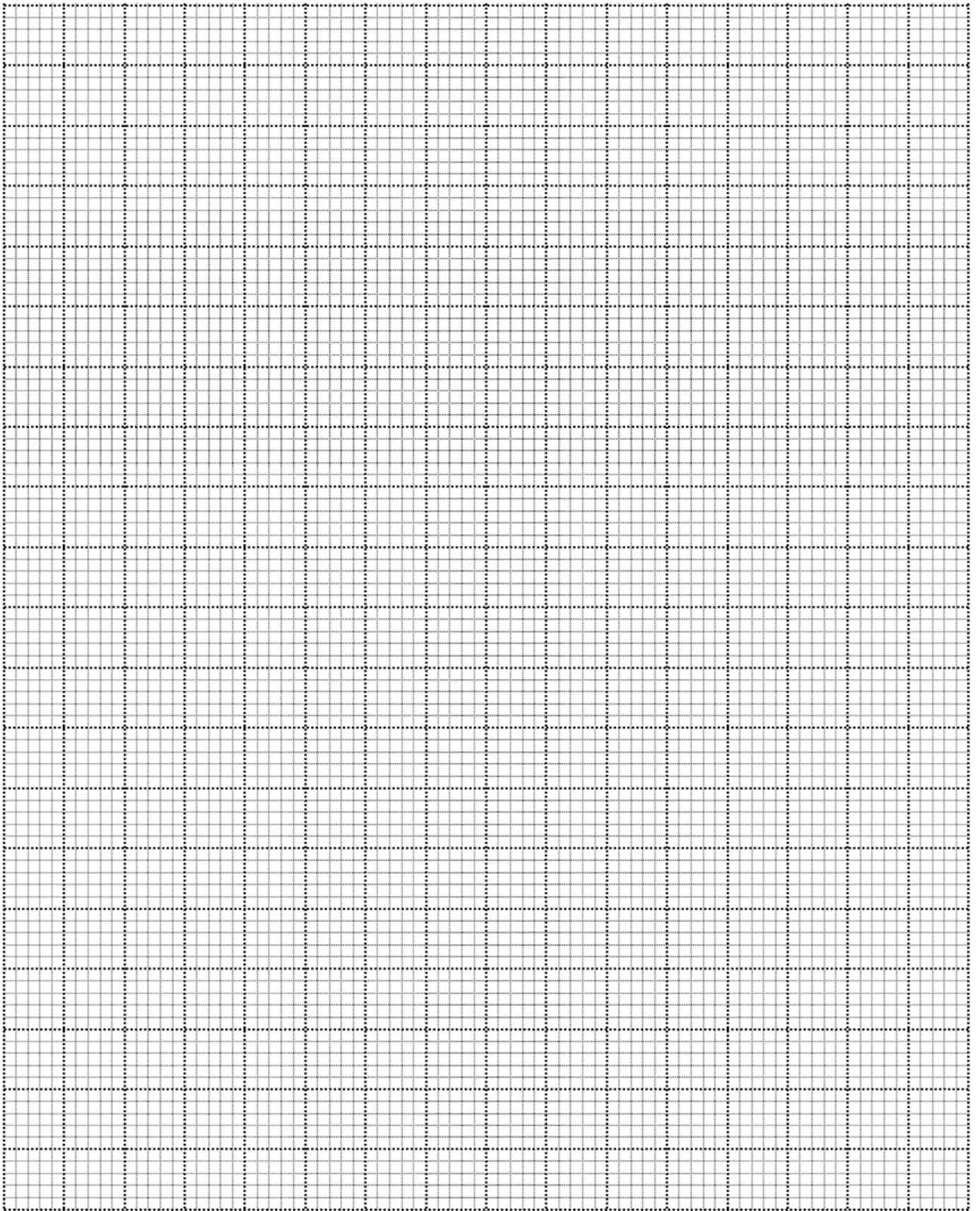
$x \text{ cm}$	1	2	3	4	5
$V \text{ cm}^3$	3.8	10.8	21.0	34.4	51.0

- (a) On the grid on the next page, plot $\frac{V}{x}$ against x and draw a straight line graph. [3]

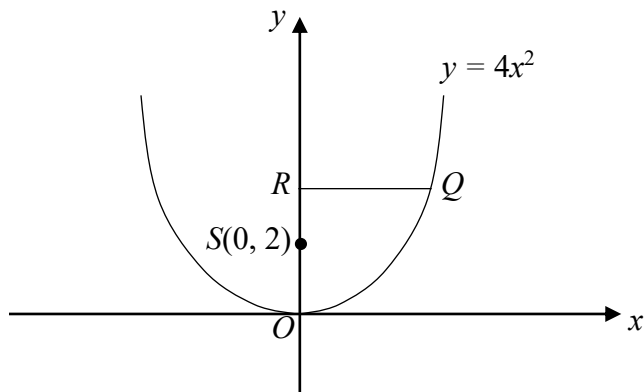
- (b) Use your graph to estimate the value of a and of b . [2]

- (c) Use your graph to estimate the height of a prism given that the volume of the prism is 6 times its height. [2]

- (d) Peter estimates the volume of the prism is 100 cm^3 when the height is 8 cm . Explain with working if Peter is correct in his estimation. [2]



- 7 The diagram below shows part of the curve $y = 4x^2$. The point R lies on the y -axis and the point Q is on the curve such that RQ is parallel to the x -axis and RQ has a length of p units.



- (a) Given that the point S is $(0, 2)$, show that the area, A , of the triangle RQS is $(2p^3 - p)$ units².

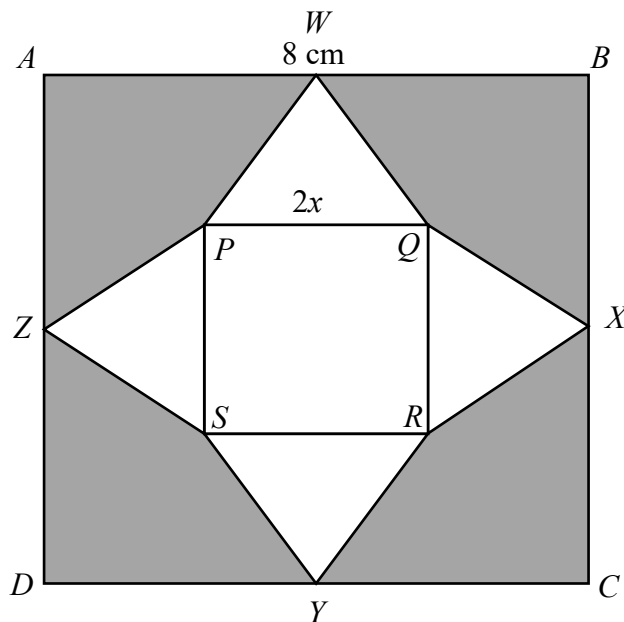
[2]

- (b) The point R moves along the y -axis and the point Q moves along the curve such that RQ remains parallel to the x -axis and the distance RQ increases at a rate of 0.03 units per second.

Find the rate of change of the area A when $p = 2$.

[3]

- 8 The diagram shows a square plastic sheet $ABCD$ of side 8 cm. W , X , Y and Z are midpoints of AB , BC , CD and AD respectively such that triangles PQW , QXR , SYR and PZS are congruent triangles. $PQRS$ is a square of length $2x$ cm. The shaded regions are to be removed and the remaining plastic is folded to form a right pyramid with base $PQRS$.



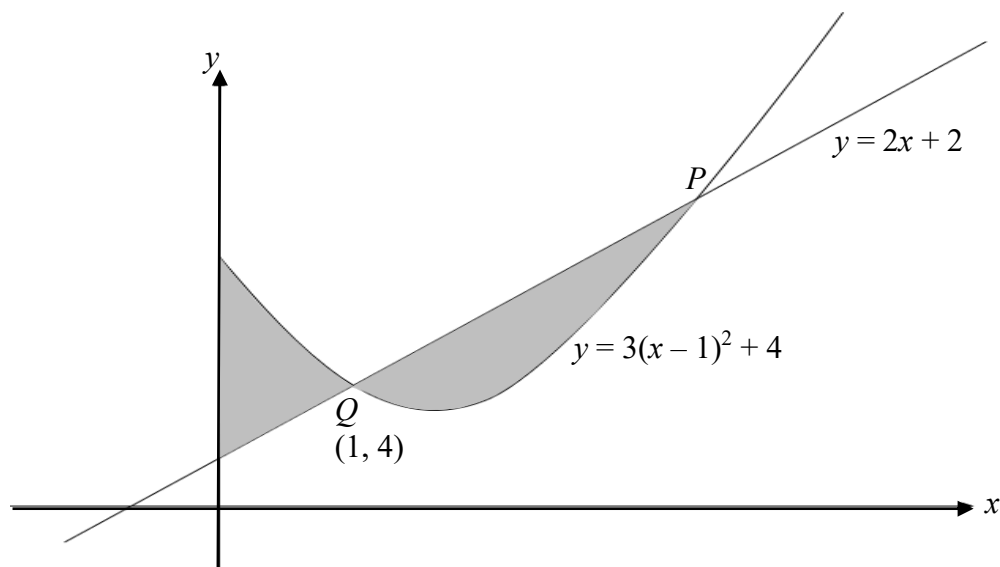
- (a) Show that the height of the pyramid is $2\sqrt{4-2x}$ cm.

[3]

- (b) Given that $V \text{ cm}^3$ is the volume of the pyramid, find the value of x which gives a stationary value of V .
Determine the nature of this stationary value.

[6]

- 9 The diagram below shows part of the curve $y = 3(x-1)^2 + 4$ intersecting the line $y = 2x + 2$ at points P and Q (1, 4).



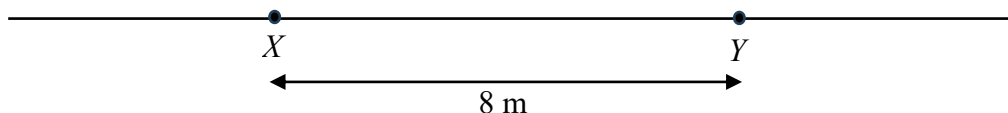
- (a) Find the coordinates of point P .

[3]

(b) Find the area of the shaded region.

[5]

10



The diagram above shows two fixed points X and Y on a straight line, where $XY = 8$ m.

A particle P moves along the line so that its velocity, v m/s, is given by $v = 2t^2 - 9t - 5$, where t is the time in seconds after leaving Y .

Initially particle P is at Y , moving towards X .

- (a) Find the acceleration of P at $t = 1$.

[2]

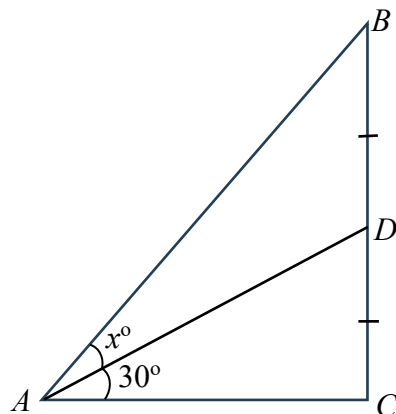
- (b) Find an expression, in terms of t , for the displacement of P from X .

[3]

- (c) Find the total distance travelled by P in the interval $t = 0$ to $t = 10$.

[4]

- 11 The diagram shows a right-angled triangle ABC and D is the midpoint of BC . Angle $DAC = 30^\circ$ and angle $BAD = x^\circ$.



- (a) Show that $AB = \frac{\sqrt{7}}{2} AD$. [3]

- (b) Show that $x^\circ = \tan^{-1}\left(\frac{\sqrt{3}}{5}\right)$. [3]

- 12** A freezer keeps the food frozen at $-10\text{ }^{\circ}\text{C}$. A piece of frozen chicken is taken out before cooking. The temperature $T\text{ }^{\circ}\text{C}$ of the chicken after t minutes of being removed from the freezer is given by $T = 32 - Ae^{-mt}$, where A and m are constants.

(a) Explain why $A = 42$. [1]

(b) Given that the temperature of the chicken rose to $20\text{ }^{\circ}\text{C}$ after an hour, find the value of m . [3]

(c) The chicken should only be cooked when the temperature of the chicken is at least $22\text{ }^{\circ}\text{C}$. Find the value of t at which the chicken is safe to be cooked. Give your answer correct to the nearest minute. [2]

(d) Explain why the temperature of the chicken can never reach $32\text{ }^{\circ}\text{C}$. [1]

End of Paper